Q July 28, 2026 — Cart on a Track

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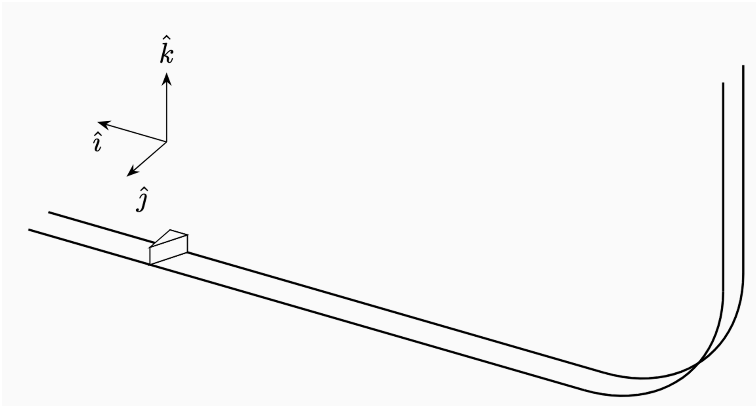
[15 pts]

A vertical track smoothly transitions to a horizontal track through a quarter-circle of radius h. A cart is released from rest at height h on the vertical track. The horizontal track is rough, with kinetic friction coefficient μ_k; the vertical and curved portions are frictionless.

On the horizontal track there is a wall, inclined at an angle θ to the track's direction (so θ is the angle between the wall's *surface* and the track, *not* the angle between the wall's normal and the track). The distance from the start of the horizontal track to the wall is L_1.



Assumptions

1. The vertical track and the quarter-circle are frictionless; only the horizontal track and the plane after the wall are rough (kinetic friction μ_k).
2. The wall is smooth and fixed, so the collision is elastic: the velocity component along the wall is unchanged, and the component normal to the wall reverses. (The cart's speed is therefore unchanged by the collision.)
3. The cart is a point mass; it never leaves the track during the descent.

Questions.

Q1. Entry speed. Using conservation of energy, find the speed v_0 of the cart just as it begins moving on the horizontal track.

Q2. Speed before the wall. Find the speed v_1 of the cart just before it reaches the wall.

Q3. After the collision. Find the x- and y-components of the velocity vector after the collision, taking $+x$ along the track and $+y$ perpendicular to it in the horizontal plane.

Q4. Sliding to rest. After the collision the cart slides on the flat plane (the same material as the horizontal track). Find the distance it covers after the collision before stopping.

Q5. Maximum acceleration. Find the maximum acceleration the cart experiences after it begins to move on the track.

Q6. Total path length. Find the total distance (path length) covered by the cart after release.

Q7. Total displacement. Find the total displacement of the cart after release (i.e. the vector from the release point to the final stopping point).